

Looped boat propellers are being tested to reduce noise that can disrupt whales

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For whales, sound is an essential source of information. Calls help them communicate, coordinate and find food. In quiet conditions, whales can hear one another's calls from hundreds of miles away. Human-made ocean noise can interfere with those sounds.

Commercial shipping produces low-**frequency** noise that overlaps with the frequencies many whales use. **Frequency** describes how rapidly a sound vibrates and is measured in hertz. Different marine species use different **frequency** ranges to communicate.

Noise can also complicate feeding. Researchers found an **association** between louder surroundings and changes in orca hunting: the animals were more likely to search for food but less likely to catch a fish they pursued. Such findings help explain the interest in quieter vessels.

An important source of propeller noise is **cavitation**. When a propeller spins through water, it can create tiny bubbles that collapse, releasing energy as sharp pressure waves. Millions of these bubbles can form and collapse each second, adding up to substantial underwater sound.

Detroit-based Sharrow Marine has developed blades that form twisting loops rather than ending in free tips. The company says this reduces tip **vortices**, the swirling flows created near blade ends, and therefore reduces **cavitation**.

Sharrow claims independent testing has shown underwater noise reductions of up to seven **decibels** in the **frequency** range whales use to communicate. **Decibels** express sound levels. The company also reports other reductions across broader **frequency** ranges, but says results vary with **frequency**, application and operating conditions. These figures are company-reported claims, rather than proof of a particular benefit to whales.

The distinction between sound above and below water matters. A boat that sounds quieter to its passengers is not automatically quieter where marine animals listen. Underwater measurements are needed to assess that question.

Sharrow Marine is partnering with Ocean Alliance to test its propellers during research expeditions near whales. The work is still early, and whether the design meaningfully improves animals' communication or feeding remains to be investigated.

Founder Greg Sharrow originally approached the problem while trying to film a violinist within an orchestra using a drone. Its buzzing interfered with recording. His search for a quieter propeller eventually shifted toward boats, with more than 640 **prototypes** developed before a partnership with the University of Michigan.

The tests near whales will explore a question that sound measurements alone cannot settle: how might quieter propellers affect the animals?

Word Treasure

frequency -- How rapidly a sound vibrates, measured in hertz.

- association** -- A link between things observed together; it does not by itself show that one causes the other.
- cavitation** -- The formation and collapse of tiny bubbles in water, which can produce sound.
- vortices** -- Swirling flows of water or air.
- decibels** -- Units used to express sound levels.
- prototypes** -- Early versions of a design made to try out and improve.

Background you need

Frequency is measured in hertz and describes how rapidly a sound vibrates. Decibels express sound levels.

Different marine species communicate using different frequency ranges.

Measurements above water and underwater answer different questions about boat noise.

Break it down

Introduces whale communication and overlapping shipping noise.

Describes the reported association between noise and orca hunting.

Explains cavitation and Sharrow Marine's proposed design solution.

Attributes noise-reduction figures to the company and explains their limits.

Describes the Ocean Alliance partnership and questions about animal benefits.

Traces the invention from drone recording to boat propeller prototypes.

Why it matters

Shipping noise can interfere with sounds whales use. Testing the propellers near whales could clarify whether reported noise reductions lead to benefits for communication or feeding.

Test what you remember

1. Why can commercial shipping noise interfere with whale communication?

- ☐ (A) It overlaps with light whales can see.
- ☐ (B) It changes the depth of ocean water.
- ☐ (C) It changes the direction of ocean currents.
- ☐ (D) It overlaps with frequencies whales use.

2. What hunting pattern did researchers associate with louder surroundings?

- ☐ (A) Less searching and more successful catches
- ☐ (B) Less searching and fewer successful catches
- ☐ (C) More searching and more successful catches
- ☐ (D) More searching and fewer successful catches

3. According to Sharrow Marine, how do its looped blades reduce cavitation?

- ☐ (A) By directing whale calls toward the surface
- ☐ (B) By reducing swirling flows near blade ends
- ☐ (C) By increasing bubble formation behind boats
- ☐ (D) By moving the propellers above the water

4. What underwater noise reduction does Sharrow claim independent tests found in the range whales use?

- ☐ (A) Up to three decibels
- ☐ (B) Up to seven decibels
- ☐ (C) Up to twenty decibels
- ☐ (D) Up to fifteen decibels

5. Why are underwater sound measurements needed?

- ☐ (A) Whales hear only the sounds made by drones.
- ☐ (B) Quieter passenger areas guarantee quieter water.
- ☐ (C) Sound frequencies cannot be measured on boats.
- ☐ (D) Quieter passenger areas may not mean quieter water.

6. What remains uncertain while tests near whales are still early?

- ☐ (A) Whether Greg first worked on quieter drones
- ☐ (B) Whether the blades form twisting loops
- ☐ (C) Whether the company works with Ocean Alliance
- ☐ (D) Whether whale communication or feeding improves

Answer key (try the quiz first!): 1: D 2: D 3: B 4: B 5: D 6: D

MY THOUGHTS (at least 20 words):